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EXAMEN CODE:

**Specific instructions for the Chemistry Contest**

Highly esteemed student:

- This exam was created by a Scientific Committee composed of former olympic students who work (mostly) at the University of Havana.
- **Complete your personal information.** Please verify with your ID card.
- The direction of the Provincial Education is responsible for ensuring that the exam is printed in a sufficiently high quality, allowing students to read the instructions, questions, figures, data, graphs, and tables. **No questions will be canceled due to poor printing.**
- Write all your information clearly in the space provided (first name, last name, school, municipality, province). **Do not write anything in the CODE field,** it is for the exclusive use of the coding manager.
- **Check that your exam is complete.** In the case of the exam on the second day, 10th grade should answer problems 1, 2, 3 and 4; 11th grade should answer problems 1, 2, 3 and 5; 12th grade should answer problems 1, 2, 5 and 6.
- All necessary constants, conversions, equations, and periodic table are included on the first page of the exam. This page will be removed during the coding process.
- At the beginning of each question, the grading is indicated, its value (out of a total of 100 points), as well as the value of each item (in marks).
- You can use a scientific calculator and a ruler.
- Please comply to the rules of approximate calculation. Attention! You have limited time and the exam has a large number of items. You do not have a space limit to answer, BUT please answer in an orderly manner, with appropriately sized and legible font, to facilitate the grading process.
- You must present all auxiliary calculations, justifying your reasoning and answers properly whenever necessary. **Unintelligible answers will not be graded.**
- Number the additional answer sheets at the bottom of each page and indicate the total number of pages, for example, “1 of 7,” “2 of 7,” and so on.
- If the exam instructor/proctor interferes with any of the above guidelines due to ignorance, exercise your right to protest and immediately request the intervention of the competent authority of the Ministry of Education, which is the institution responsible for administering the exam.
- The President of the Scientific Committee can be reached at +53-5045-4299.

You are a source of pride for Cuban education: Best of luck!

EXAMEN CODE (only for the use of the code manager)

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal Gas Law	$p \cdot V = n \cdot R \cdot T$
Gas constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero in Celsius scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pression lowering	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Sphere volume	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Sphere area	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Circle area	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of a circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Gravity force	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

EXAMEN CODE (only for the use of the code manager)

Problem 7. Carbon and sulfur (15% of total; 10th, 11th and 12th grades)

7.1	7.2	7.3	7.4	7.5	7.6	Total	Total
10	8	4	8	2	28	60	15

Ernesto is a very curious student, and whenever he has the opportunity, he conducts experiments in the school laboratory with technician Ivis and mentor Agustín. After conducting a bibliographic search, the mentor discovered that when carbon and sulfur combine, different binary substances are obtained, and he asked the student to answer the following question.

7.1. What is the formula for the four sulfur compounds of carbon containing approximately 36, 27, and 16% carbon by mass? It is known that in one case, both the monomer and the dimer can exist, and none of the molecules is made up of more than five atoms.

7.2. Help Ernesto to represent the structures of the four molecules described above.

Ernesto liked the challenge and asked permission to try to obtain the substances at the school. But Ivis denied him, arguing that sulfur compounds are toxic and that the reactions involved were highly exothermic. Fortunately, Agustín has excellent relations with the University of Sancti Spiritus, and they planned to perform the reactions under more controlled conditions the following day.

Upon arriving at the university, Ivis found a **10,0 L** rigid reactor equipped with a pressure gauge and placed it under a fume hood. Ernesto weighed 60,0 g of a carbon-sulfur mixture on a technical balance and placed it inside the container. At that moment, Agustín sent them out for a snack, and when he was left alone in the laboratory, he secretly modified the experimental conditions. Upon returning, the student pressed a button, producing a spark and a small explosion inside the reactor. After cooling to room temperature, Ernesto was shocked and disappointed because the pressure inside the container was back to where it had been. Upon careful opening, he noticed a very unpleasant odor and the absence of solid residue. Then the mentor confessed: he had removed most of the solid mixture and filled the container with oxygen.

7.3. Can you help Ernesto to formulate the chemical reactions that happened?

“Agustín, you’re always so reckless!” Ivis said angrily and expelled him from the room. Ernesto weighed the portion of the carbon-sulfur mixture his mentor had hidden. He found a mass of only **57,3 g** and poured it completely out of the reactor. The technician made sure to evacuate all the air and replace it this time with the inert gas argon, until a pressure of 1,00 atm was reached. A spark was then ignited, and a strongly exothermic reaction occurred inside the vessel, resulting in the formation of substances **A** and **B** in a gaseous state. No solid residue was observed through the glass window of the apparatus. The system was then placed in an ice bath. After cooling to 100 °C, the pressure inside the vessel was 3,81 atm. Cooling continued, and upon reaching the temperature of 25 °C, a few drops appeared on the window due to the condensation of some of product **B**, and the pressure dropped to 1,96 atm. Upon reaching 0 °C and after carefully evacuating the gases (only Ar and product **A**), between 37 and 38 mL of a clear, volatile liquid with a very unpleasant odor were recovered (all of product **B**), which, when measured with a hydrometer, had a density of 1,29 g/mL.

7.4. Show that the amount of oxygen initially introduced into the reactor by the mentor was sufficient to react with the small portion of the mixture that was combusted.

Vapor pressure is the pressure exerted by the portion of a volatile substance in the gaseous phase, in direct contact –that is, in equilibrium– with the corresponding volatile liquid or solid. Vapor is said to be “saturated” when the gaseous phase contains the largest amount of the volatile substance thermodynamically possible.

7.5. The saturated vapor pressure of **B** at 25 °C is 48,1 kPa. Calculate the maximum amount of **B** in the gas phase that the reactor can hold at that temperature.

7.6. Help the student to calculate the identity of **A** and **B**.

Note: The experiments described in this problem are entirely fictitious and extremely dangerous. **Never** attempt to reproduce them.

Problem 8. Benzoyl peroxide (10% of total; only 10th and 11th grades)

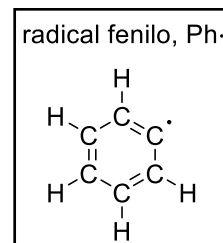
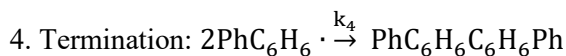
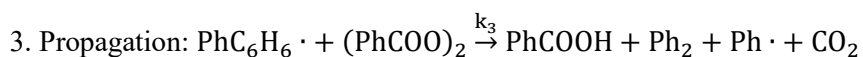
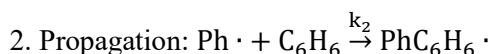
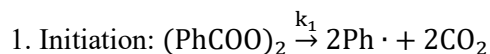
8.1	8.2	8.3	8.4	8.5	8.6	8.7	Total	Total
2	10	4	10	2	4	8	40	10

Organic peroxides decompose easily, generating free radicals, which is why they are used as initiators in chain reaction, especially in industrial polymerization processes. One day, Bárbaro and Pedro studied the decomposition of benzoyl peroxide $(\text{PhCOO})_2$ by refluxing in the solvent benzene. These students work well as a team, as their skills complement each other: the former has solid theoretical knowledge, and the latter is very capable in the laboratory.

Pedro's measurements demonstrated that the rate law is first-order with respect to peroxide. However, the values of the reaction rate constants k_{obs} were found to be dependent on the initial concentration of benzoyl peroxide $[\text{PB}]_0$, as shown in the data in the following table.

$[\text{PB}]_0$ (mol/L)	0,0205	0,0830	0,156	0,225	0,283
k_{obs} (10^{-5} s^{-1})	3,85	5,30	6,35	7,10	7,70

These results suggested that, in addition to the unimolecular thermal decomposition of benzoyl peroxide, decomposition processes mediated by some free radical intermediate also occur. Bárbaro found that the following simplified decomposition mechanism is reported in the literature.



8.1. Why is the O-O single bond present in peroxides relatively weak?

Bárbaro applied the Steady State Theory to the mechanism and showed that the rates of the initiation step 1 and termination step 4 of the mechanism have to be equal, and that the rate of decomposition of the peroxide obeys a complex expression of the form $v_r = -\frac{d[\text{PB}]}{dt} = (\mathbf{A} + \mathbf{B} \cdot \sqrt{[\text{PB}]}) \cdot [\text{PB}]$, where the constants **a** and **b** can be expressed in terms of the specific rate constants of each step.

8.2. Deduce the complete reasoning that Barbaro followed.

8.3. The values of the specific rate constants are $k_1 = 2,4 \cdot 10^{-5} \text{ s}^{-1}$ and $k_4 = 2,0 \cdot 10^{11} \text{ u}$. State the units of measurement “u” of k_4 and calculate the concentration of the radical $\text{PhC}_6\text{H}_6 \cdot$ if we start from an initial peroxide concentration of 0,010 mol/L.

8.4. Fit Pedro's results in the table to a straight line and calculate the values of the constants **A** and **B** of Barbaro's rate equation

$[\text{PB}]_0$ (mol/L)	0,0205	0,0830	0,156	0,225	0,283
k_{obs} (10^{-5} s^{-1})	3,85	5,30	6,35	7,10	7,70

8.5. When Pedro asked Bárbaro why the fourth reaction step (termination) is the fastest of all, what do you think his answer was?

8.6. At low concentrations of peroxide $k_{\text{obs}} = k_1$, while at high concentrations of peroxide $k_{\text{obs}} = k_3 \sqrt{\frac{k_1}{k_4}}$. Use the

Arrhenius model and express the experimentally observed activation energy for the reaction E_A^{obs} as a function of the activation energies of the individual steps for both cases.

8.7. Estimate the minimum values of the activation energies E_{A1} , E_{A2} , E_{A3} and E_{A4} .

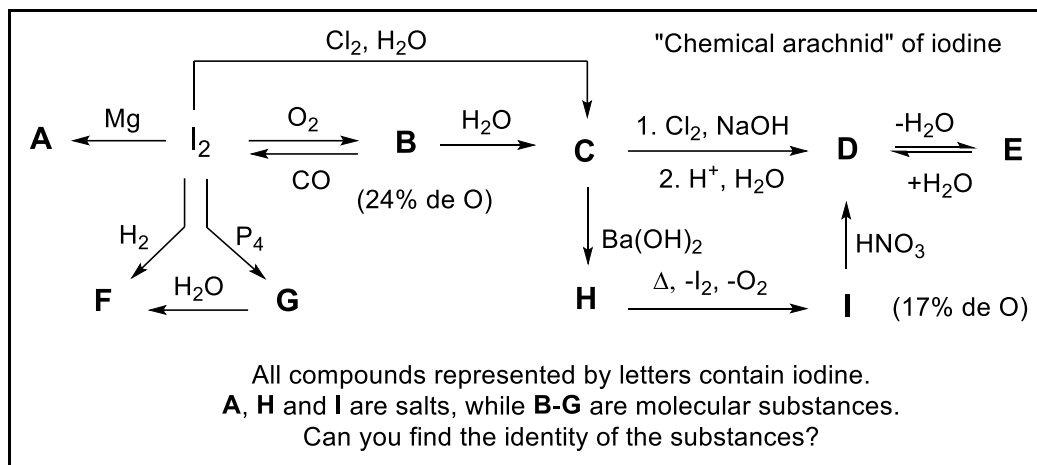
enlace	H-H	C-C	O-O	C-H	C-O	O-H	C=C	C=O
E (kJ/mol)	436	348	146	413	358	463	612	799

Problem 9. Iodine (15% of total; only 10th and 11th grades)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	Total	Total
20	11	7	2	2	5	4	4	5	60	15

When Melissa discovered that Cynthia was afraid of spiders, she came up with a plan to help her overcome her fears. She took a piece of paper that was black on one side and white on the other, wrote a note on the white side, and folded it to create an origami arachnid. Melissa then pretended to be holding a real spider, and Cynthia began to scream in terror –until Melissa confessed–, “Hold it. It won’t

bite you; it’s a chemical spider.” After gathering a lot of courage, Cynthia took the origami paper and found the following problem written on the white side of the paper sheet.



9.1. Help Cynthia to find the formulas for substances A-I. Record your calculations.

Hint: D and E are iodine oxoacids in the same oxidation state. The molecule of D has a square-base bipyramid geometry, in which five I-O bonds measure 1,87-1,91 Å, and one I-O bond measures 1,78 Å.

9.2. Draw the structures of the iodine species shown in Melissa’s note. Note the geometry according to the postulates of Gillespie’s theory (TRPEV).

9.3. Melissa discovered in an article that the reaction between diiodine and A in concentrated sulfuric acid is complex and results in the formation of the species IO⁺, I₂O, I₃⁺, I₅⁺. Draw their structures.

9.4. At room temperature, iodine is solid, while bromine is liquid, but chlorine and fluorine are gases. Why?

9.5. Hydrogen iodide behaves like a strong acid when dissolved in water, completely dissociating into ions. In contrast, hydrogen fluoride behaves like a weak acid and only partially dissociates. Why is this happening?

9.6. Describe the diiodine molecule using Molecular Orbital Theory and state: i) electron distribution, ii) bond order, iii) magnetic properties, and iv) frontier orbitals.

9.7. Iodine reacts with the sulfite anion to form sulfate and iodide ions. Propose a probable mechanism for the reaction, indicating the movement of electrons with arrows.

On the day of the science fair, Cynthia decided it was time for revenge. While Melissa was studying unsuspectingly, Cynthia threw some ammonia and iodine in the hallway, right outside the lab entrance, and waited patiently. When the break time bell rang, explosions and screams were heard. It was Melissa, scared, not understanding why the floor was exploding beneath her shoes. And the rest of the chemistry group was laughing out loud, well aware of their mentors’ favorite joke. The explosive compound that forms *in situ* during a chemical reaction has the formula NI₃·NH₃.

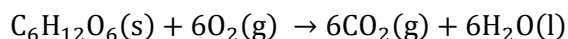
9.8. Propose a probable mechanism for the reaction of nitrogen triiodide formation from ammonia and diiodine, indicating the movement of electrons with arrows.

9.9. Draw the structure of nitrogen triiodide ammonate. Explain why this compound is extremely unstable and detonates at the slightest vibration or contact. Write the equation for the decomposition reaction that occurs during the explosion.

Problem 10. Hiking in Gran Piedra (15% of total; only 10th grade)

10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	Total	Total
9	6	4	11	6	4	10	10	60	15

Mathematician Ian and physicist Sergio accompanied the chemists who climbed the Gran Piedra (1,225 m above sea level), at the eastern end of the Sierra Maestra mountain range, in Santiago de Cuba, a few weeks ago. The total distance traveled was approximately 30 km, requiring a large expenditure of energy –approximately 2,500 calories. The day before leaving, Ian decided to calculate the amount of food he should take. To do so, he calculated the heat released during the combustion of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), which occurs at the cellular level and is the main source of energy for the human body:



10.1. How much food (grams of glucose) does a student need to complete the hike?

10.2. State three reasons why Ian's result is unreliable.

On the day of the tour, a meteorological phenomenon known as a trough was over eastern Cuba, causing southwesterly winds loaded with moisture from the Caribbean Sea. As a result, as they approached the summit, just over a kilometer high, the students entered a cloud that remained covering the area all day. Sergio asked the tour guides about the weather and learned that as they left Santiago, on the seashore, the temperature was 1,000 hPa and 25 °C, while at Gran Piedra, both values had dropped to 990 hPa and 15 °C, respectively.

10.3. Consider a 1,00 km³ air mass from the Caribbean Sea moving over Santiago de Cuba toward the mountains. Assuming that there is no diffusion and no mixing with the surrounding air, calculate the volume of the air mass as it ascends to Gran Piedra. Consider ideal behavior.

If you cannot solve item **10.3**, consider that the volume of the air mass remains 1,00 km³.

Meteorologists use the variable relative humidity (H_r), which is the ratio of the actual pressure exerted by gaseous water (vapor) to the vapor pressure saturating the air at a given temperature. When the air is saturated with water vapor, relative humidity is $H_r = 100\%$. During his time inside the cloud, Sergio wanted to know how much water had condensed in the form of cloud, rain, and dew, as a result of the temperature change. To do this, he used water vapor pressure values (1,70 kPa at 15 °C and 3,17 kPa at 25 °C) and assumed that the air above Gran Piedra was saturated with water vapor, meaning that it contained the maximum possible amount of water in a gaseous state.

10.4. Estimate the maximum amount (L) of precipitation that can be generated from the air mass of 1,00 km³ coming from the Caribbean Sea if it is known that over the city of Santiago de Cuba $H_r = 85\%$.

If you cannot solve item **10.4**, consider that 5,00 million liters of rainwater fall.

10.5. How much heat (in kJ) was released into the atmosphere as a result of the condensation of water?

During the excursion, Ian learned that acid rain is common on Gran Piedra. This is because the trade winds carry fumes from the Moa industrial zone. This polluted air contains sulfur dioxide and nitric oxide (NO), which, when in contact with oxygen and humidity, generate highly corrosive sulfuric and nitric acids.

10.6. Write the equations corresponding to the formation of acid rain.

On the way back, Sergio picked up a shiny black stone he found along the way. Upon returning to school and examining the stone, he accidentally discovered that it was attracted to magnets and remained magnetized after the process. He then asked the chemists to analyze the material in the laboratory of the Training Center. The sample dissolved slowly in hot, concentrated hydrochloric acid, leaving no solid residue and creating a yellow-green solution. A few drops of the solution were then added to test tubes containing 2 cm³ of 0,5 M solutions of potassium rhodanide (sulfocyanide, thiocyanate, KSCN), sodium hydroxide, and ammonia. In the first case, the solution turned deep red, while in the other two, precipitates appeared that changed color until they turned the same brown. The coordinator, Rafael, who has very good

connections, requested a quantitative instrumental analysis using ICP and XPS in a specialized laboratory of the Geominera Oriente company. The stone was found to be composed of a single metal oxide with the formula M_3O_4 .

10.7. Identify the metal contained in the stone and write the equations for all the reactions that occurred during the qualitative analysis.

10.8. Calculate the lattice energy of the mixed oxide M_3O_4 from the thermochemical data given. Hint: Note that both M^{2+} and M^{3+} ions are present in the solid.

$1 \text{ cal} = 4,184 \text{ J}$ $\rho(\text{H}_2\text{O}) = 1,00 \text{ g/mL}$ $1 \text{ bar} = 100 \text{ kPa} = 1000 \text{ hPa} = 0,987 \text{ atm}$ $1 \text{ L} = 1 \text{ dm}^3$

ΔH_f° (kJ/mol, 25°C): $\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) = -1273$, $\text{CO}_2(\text{g}) = -394$, $\text{H}_2\text{O}(\text{l}) = -286$, $\text{H}_2\text{O}(\text{g}) = -242$, $\text{M}_3\text{O}_4 = -1117$

$D(\text{O}_2) = 498 \text{ kJ/mol}$, $A_1(\text{O}) = -141 \text{ kJ/mol}$, $A_2(\text{O}) = 845 \text{ kJ/mol}$, $I_1(\text{M}) = 759 \text{ kJ/mol}$, $I_2(\text{M}) = 1561 \text{ kJ/mol}$, $I_3(\text{M}) = 2957 \text{ kJ/mol}$; $\Delta H_{\text{atomization}}^\circ(\text{M}) = 415 \text{ kJ/mol}$.

ANSWER FROM HERE ON
